Sahaj Bhadja

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Summary

Software engineer with 2+ years building **validation-oriented tooling** in **C/C++ and Python**, automating **Linux/Windows workflows**, and using AI-assisted development (Copilot/Claude-style prompting) to improve delivery speed and code quality.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **Python-based validation services** and interfaces for 500+ users, reducing workflow latency from **1.2s to 350ms** through profiling, query tuning, and structured test checks.
- Implemented **fault-tolerant orchestration** for 5K+ monthly jobs with validation safeguards and rollback logic, improving reliability during rapid feature enablement cycles.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **Flask/React testable modules** handling 10K+ daily requests, converting requirements into reusable components and backend APIs for production-grade validation workflows.
- Improved **CI/CD automation and debug loops** using script-driven checks and issue triage, reducing release turnaround by **60%** while maintaining quality targets.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **large-scale experiment pipelines** for 100K+ trajectories, enabling reproducible OS-level workload analysis and benchmark reporting for complex model/system behavior.
- Used **LLM-assisted summarization and debugging** to accelerate documentation and root-cause tracking, improving handoff quality between research and engineering collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **backend integration services** with Python/Java, AWS Glue, and S3, improving response time by **30%** and reliability for distributed processing pipelines.
- Maintained **20+ automated unit/integration suites** with Jenkins, PyTest, and Mockito on Linux/Windows, increasing pipeline efficiency by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Implemented **MERN + WebSocket data modules** supporting eight workflows and 2M+ synchronized records, improving correctness and concurrency in high-frequency operations.
- Automated **containerized deployment/testing** through Docker/Jenkins workflows, reducing rollout time by **50%** and improving repeatability across environments.

Skills

Languages: **C/C++**, **Python**, Java, JavaScript/TypeScript, SQL, Bash
Systems: Linux/UNIX, Windows tooling, OS fundamentals, computer architecture basics, multithreading
Validation & Debug: feature validation content, troubleshooting, root-cause analysis, reproducible benchmarking
AI-Assisted Dev: Copilot/Claude-style prompt workflows, code generation review, GenAI safety-aware usage
Tooling: Visual Studio Code, Git/GitHub, Jenkins, CI/CD, Make/CMake, Agile/Scrum documentation

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented **FCFS, SJF, SRT, RR, and MLFQ** with timing metrics and trace visualizations, reducing debugging time by **20%** via deterministic, test-driven scheduling analysis.

Synthetic Workload Generator | *C++, Python, Linux, JSON*
- Built a **config-driven workload generator** to produce validation scenarios across compute and I/O patterns, improving test coverage and repeatability for system feature verification.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized REST pipeline** on AWS with automated CI/CD and validation checks, improving production reliability for sustained operational workloads.